EDHEC BUSINESS SCHOOL

Lecture 9
Correlation, Regression

APPENDIX

Beta

June 2026

UNLEASH TOMORROW

Recap from Part 1: covariance, correlation, and slope

From Lecture 9 Part 1, remember that:

- Correlation measures linear association after standardizing units;
- Covariance measures co-movement in the original units.

$$\mathrm{Cov}(X,Y) = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})$$

$$r_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{s_X s_Y}$$

And that in simple linear regression, the slope can be written using covariance.

$$b_1 = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)}$$

$$b_1 = r_{X,Y}\frac{s_Y}{s_X}$$

Quantity	What it tells us
Covariance	direction of co-movement, with units
Correlation	standardized strength of linear association
Slope	average change in Y for one unit of X

RETURNS AND THE RISK-FREE RATE

Before introducing beta, we must first define the following finance notation.

ALL RETURNS MUST BE MEASURED OVER THE SAME PERIOD.

Symbol	Meaning
$R_{i,t}$	return on asset i in period t
$R_{m,t}$	return on the market portfolio in period t
$R_{f,t}$	risk-free return in period t
$R^e_{i,t}$	asset excess return: $R_{i,t} - R_{f,t}$
$R^e_{m,t}$	market excess return: $R_{m,t} - R_{f,t}$

$$R^e_{i,t} = R_{i,t} - R_{f,t}$$

$$R^e_{m,t} = R_{m,t} - R_{f,t}$$

Excess return is performance above the risk-free benchmark.

Finance beta is defined using excess returns, not raw returns.

FINANCE BETA IS A REGRESSION SLOPE

$$R^e_{i,t} = \alpha_i + \beta_i R^e_{m,t} + \varepsilon_{i,t}$$

This is a statistical model: it describes an average relationship, not an exact rule for every future return.

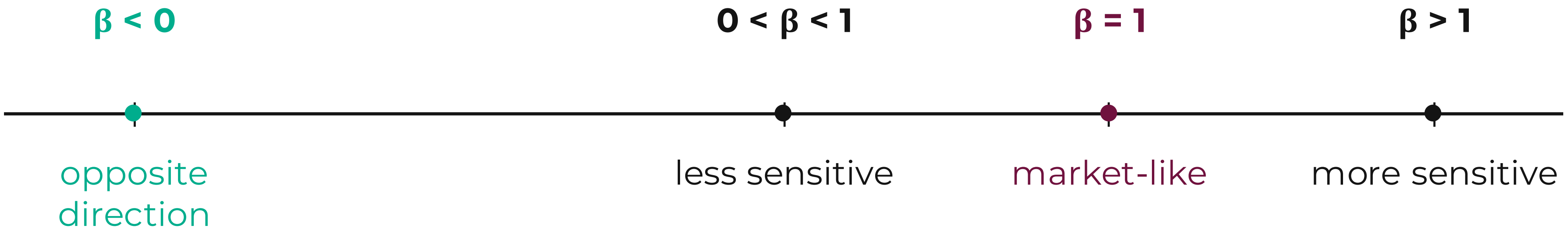
Beta is the slope from regressing asset excess return on market excess return.

The fitted line describes average exposure to market-wide movements.

Symbol	Meaning
α_i	intercept; average excess return not explained by market exposure
β_i	beta; sensitivity to market excess return
$\varepsilon_{i,t}$	residual; part not explained by the market

READING BETA

Beta is read like a slope: the average change in asset excess return when market excess return increases by one percentage point.



The sign gives the direction of exposure. The magnitude gives how strongly the asset responds to market excess returns.

BETA FROM COVARIANCE

$$\beta_i = \frac{\mathrm{Cov}\left(R_i^e, R_m^e\right)}{\mathrm{Var}\left(R_m^e\right)}$$

The numerator measures how the asset excess return moves with the market excess return.

The denominator rescales by the variability of the market excess return.

This is the same slope as the beta regression.

Interpretation

- High covariance with the market increases beta.
- Higher market variance reduces beta for a given covariance.
- Beta is about sensitivity to market excess returns.

BETA MEASURES SYSTEMATIC RISK

Beta measures exposure to market-wide movements in excess returns. It does not measure every source of risk.

Systematic market risk (captured by beta)	Idiosyncratic risk (left in the residual)

Company news, sector events, liquidity, and asset-specific shocks can still move an asset.

In the beta regression, those movements appear in the residual term ε.

ALPHA IS THE INTERCEPT, NOT AUTOMATIC SKILL

In the beta regression, alpha is the intercept.

It is the average excess return not explained by exposure to the market excess return.

$$R^e_{i,t} = \alpha_i + \beta_i R^e_{m,t} + \varepsilon_{i,t}$$

In investment language, alpha is often used to describe performance beyond market exposure.

That interpretation requires evidence. A positive estimated alpha may reflect noise, missing risk factors, fees, or a short sample.

Alpha is a claim to test, not something to assume.

ESTIMATING AND USING BETA

```
file = "Lecture 09 teaching returns.csv"
r = pd.read_csv(file)

rf = 0.0
r["asset_e"] = r["asset_a"] - rf

r["market_e"] = r["market"] - rf
X = sm.add_constant(r["market_e"])
model = sm.OLS(r["asset_e"], X).fit()
beta = model.params["market_e"]
```

Python steps

1. Read the return file.
2. Compute asset and market excess returns.
3. Regress asset_e on market_e.
4. Read beta from the market_e coefficient.

What to report

- beta estimate
- standard error or confidence interval
- R-squared
- residual checks

Interpret beta as a historical estimate of market exposure, not as a mechanical rule for future returns.